

CHAPTER 5

Problems with the Predictors

5.1 Errors in the Predictors

The regression model $Y = X\beta + \varepsilon$ allows for Y being measured with error by having the ε term, but what if the X is measured with error? In other words, what if the X we see is not the X used to generate Y ? It is not unreasonable that there might be errors in measuring X . For example, consider the problem of determining the effects of being exposed to a potentially hazardous substance such as secondhand tobacco smoke. Such exposure would be a predictor in such a study, but clearly it is very hard to measure this exactly over a period of years.

One should not confuse the errors in predictors with treating X as a random variable. For observational data, X could be regarded as a random variable, but the regression inference proceeds conditional on a fixed value for X . We make the assumption that the Y is generated conditional on the fixed value of X . Contrast this with the errors in predictors case where the X we see is not the X that was used to generate the Y .

Suppose that what we observe is (x_i^O, y_i^O) for $i=1, \dots, n$ which are related to the true values (x_i^A, y_i^A)

$$\begin{aligned} y_i^O &= y_i^A + \varepsilon_i \\ x_i^O &= x_i^A + \delta_i \end{aligned}$$

where the errors ε and δ are independent. The situation is depicted in Figure 5.1. The true underlying relationship is:

$$y_i^A = \beta_0 + \beta_1 x_i^A$$

but we only see (x_i^O, y_i^O) Putting it together, we get:

$$y_i^O = \beta_0 + \beta_1 x_i^O + (\varepsilon_i - \beta_1 \delta_i)$$

Suppose we use least squares to estimate β_0 and β_1 . Let's assume $E\varepsilon_i = E\delta_i = 0$ and that $\text{var } \varepsilon_i = \sigma_\varepsilon^2$, $\text{var } \delta_i = \sigma_\delta^2$ Let:

$$\sigma_x^2 = \sum (x_i^A - \bar{x}^A)^2 / n \quad \sigma_{x\delta} = \text{cov}(x^A, \delta)$$

For observational data, σ_x^2 is (almost) the sample variance of X^A while for a controlled experiment we can view it as just a numerical measure of the spread of the design. A similar distinction should be made for $cov(x^A, \delta)$ although in many cases, it will be reasonable to assume that this is zero.

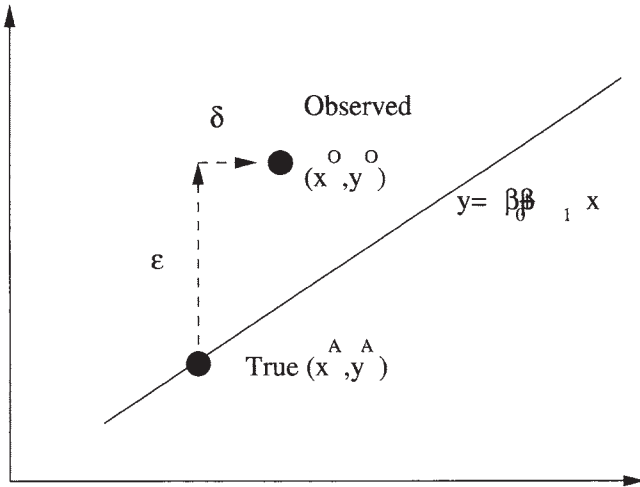


Figure 5.1 Measurement error: True vs. observed data.

Now $\hat{\beta}_1 = \sum(x_i - \bar{x})y_i / \sum(x_i - \bar{x})^2$ and after some calculation we find that:

$$E\hat{\beta}_1 = \beta_1 \frac{(\sigma_x^2 + \sigma_{x\delta})}{(\sigma_x^2 + \sigma_\delta^2 + 2\sigma_{x\delta})}$$

There are two main special cases of interest:

1. If there is no relation between X^A and δ , $\sigma_{x\delta} = 0$, this simplifies to:

$$E\hat{\beta}_1 = \beta_1 \frac{1}{1 + \sigma_\delta^2 / \sigma_x^2}$$

So $\hat{\beta}_1$ will be biased towards zero, regardless of the sample size. If σ_δ^2 is small relative to σ_x^2 then the problem can be ignored. In other words, if the variability in the errors of observation of X are small relative to the range of X , then we need not be too concerned. For multiple predictors, the usual effect of measurement errors is also to bias the $\hat{\beta}$ in the direction of zero.

2. In controlled experiments, we need to distinguish two ways in which error in x may arise. In the first case, we measure x so although the true value is x^A we observe x^0 . If we were to repeat the measurement, we would have the same x^A but a different x^0 . In the second case, you fix x^0 —for example, you make up a chemical solution with a specified concentration x^0 . The true concentration would be x^A . Now if you were to repeat this, you would get the same x^0 , but the x^A would be different. In this latter case we have:

$$\sigma_{x\delta} = \text{cov}(X^0 - \delta, \delta) = -\sigma_\delta^2$$

and then we would have $E\hat{\beta}_1 = \beta_1$. So our estimate would be unbiased. This seems paradoxical, until you notice that the second case effectively reverses the roles of x^A and x^0 and if you get to observe the true X , then you will get an unbiased estimate of β_1 . See Berkson (1950) for a discussion of this.

If the model is used for prediction purposes, we can make the same argument as in the second case above. In repeated “experiments,” the value of x at which the prediction is to be made will be fixed, even though these may represent different underlying “true” values of x .

In cases where the error in X can simply not be ignored, we should consider alternatives to the least squares estimation of β . The least squares regression equation can be written as:

$$\frac{y - \bar{y}}{SD_y} = r \frac{(x - \bar{x})}{SD_x}$$

so that $\hat{\beta}_1 = rSD_y/SD_x$. Note that if we reverse the roles of x and y , we do not get the same regression equation. Since we have errors in both x and y in our problem, we might argue that neither one, in particular, deserves the role of response or predictor and so the equation should be the same either way. One way to achieve this is to set

$\hat{\beta}_1 = SD_y/SD_x$. This is known as the *geometric mean functional relationship*. More on this can be found in Draper and Smith (1998). Another approach is to use the SIMEX method of Cook and Stefanski (1994), which we illustrate below.

Consider some data on the speed and stopping distances of cars in the 1920s. We plot the data, as seen in Figure 5.2, and fit a linear model:

```
> data (cars)
> plot (dist ~ speed, cars, ylab="distance")
> g <- lm (dist ~ speed, cars)
> summary (g)
Coefficients:
      Estimate Std. Error t value Pr(>|t|)
```

```
(Intercept) -17.579      6.758    -2.60     0.012
speed        3.932      0.416     9.46    1.5e-12
Residual standard error: 15.4 on 48 degrees of freedom
Multiple R-Squared: 0.651, Adjusted R-squared: 0.644
F-statistic: 89.6 on 1 and 48 DF, p-value: 1.49e-12
> abline (g)
```

We could explore transformations and diagnostics for these data, but we will just focus on the measurement error issue. Now we investigate the effect of adding measurement error to the predictor. We plot the modified fits in Figure 5.2:

```
> gel <- lm (dist ~ I (speed+rnorm (50)), cars)
> coef (gel)
      (Intercept)      I (speed + rnorm(50))
      -15.0619          3.7582
> abline (gel, lty=2)
> ge2 <- lm (dist ~ I (speed+2*rnorm (50)), cars)
> coef (ge2)
      (Intercept)      I (speed + 2 * rnorm(50))
      -5.3503          3.1676
> abline (ge2, lty=3)
```

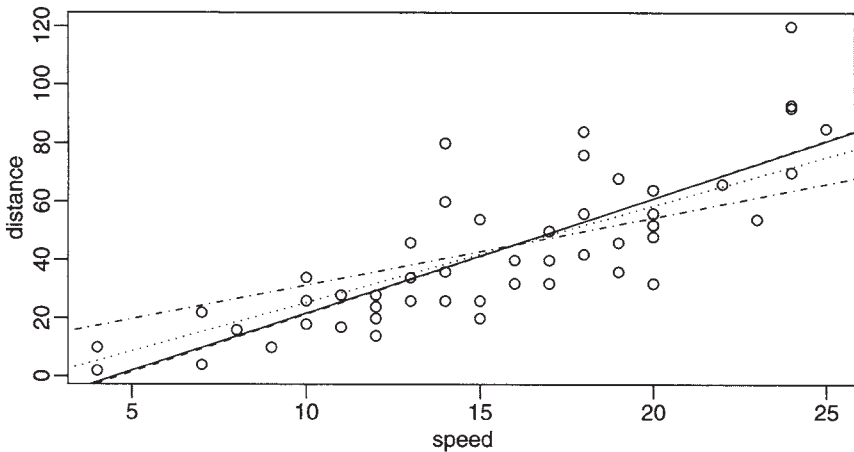


Figure 5.2 *Stopping distance and speeds of cars. The least squares fit is shown as a solid line. The fits with three progressively larger amounts of measurement error on the speed are shown as dotted lines, where the slope gets shallower as the error increases.*

```
> ge5 <- lm (dist ~ I (speed+5*rnorm(50)), cars)
> coef (ge5)
      (Intercept)      I (speed + 5 * rnorm(50))
      15.1589          1.8696
> abline (ge5, lty=4)
```

We can see that the slope becomes shallower as the amount of noise increases.

Suppose we knew that the predictor, speed, in the original data had been measured with a known error variance, say 0.5. Given what we have seen in the simulated measurement error models, we might extrapolate back to suggest an estimate of the slope under no measurement error. This is the idea behind SIMEX.

Here we simulate the effects of adding normal random error with variances ranging from 0.1 to 0.5, replicating the experiment 1000 times for each setting:

```
> vv <- rep(1:5/10, each=1000)
> slopes <- numeric(5000)
> for (i in 1:5000) slopes[i] <- lm(dist ~
  I(speed + sqrt(vv[i]) * rnorm(50)), cars)$coef[2]
```

Now plot the mean slopes for each variance. We are assuming that the data have variance 0.5 so the extra variance is added to this:

```
> betas <- c(coef(g)[2], colMeans(matrix(slopes,
  nrow=1000)))
> variances <- c(0, 1:5/10) + 0.5
> plot(variances, betas, xlim=c(0, 1), ylim=c(3.86, 4))
```

We fit a linear model and extrapolate to zero variance:

```
> gv <- lm(betas ~ variances)
> coef(gv)
```

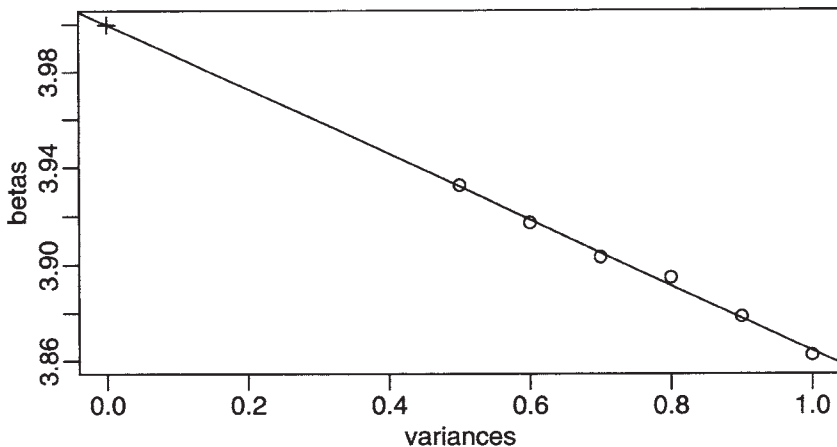


Figure 5.3 *Simulation-Extrapolation estimation of the unbiased slope in the presence of measurement error in the predictors. We predict $\hat{\beta} = 4.0$ at a variance of zero.*

```
(Intercept)  variances
3.99975      -0.13552
> points(0, gv$coef[1], pch=3)
```

The predicted value of $\hat{\beta}$ at variance equal to zero, that is no measurement error, is 4.0. Better models for extrapolation are worth considering; see Cook and Stefanski (1994) for details.

5.2 Changes of Scale

Suppose we reexpress x_i as $(x_i+a)/b$. We might want to do this because predictors of similar magnitude result in $\hat{\beta}$ of similar sizes. $\hat{\beta} = 3.51$ is easier to parse than $\hat{\beta} = 0.000000351$ and we can choose a and b to achieve this. Furthermore, a change of units might aid interpretability. Finally, numerical stability in estimation is enhanced when all the predictors are on a similar scale.

Rescaling x_i leaves the t - and F -tests and $\hat{\sigma}^2$ and R^2 unchanged and $\hat{\beta}_i \rightarrow b\hat{\beta}_i$

Rescaling y in the same way leaves the t - and F -tests and R^2 unchanged, but $\hat{\sigma}$ and $\hat{\beta}$ will be rescaled by b .

To demonstrate this, we use the same old model:

```
> data(savings)
> g <- lm(sr ~ pop15+pop75+dpi+ddpi, savings)
> summary(g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  28.566087   7.354516   3.88  0.00033
pop15        -0.461193   0.144642  -3.19  0.00260
pop75        -1.691498   1.083599  -1.56  0.12553
dpi          -0.000337   0.000931  -0.36  0.71917
ddpi         0.409695   0.196197   2.09  0.04247
Residual standard error: 3.8 on 45 degrees of freedom
Multiple R-Squared:  0.338, Adjusted R-squared:  0.28
F-statistic: 5.76 on 4 and 45 DF, p-value: 0.00079
```

The coefficient for income is rather small—let’s measure income in thousands of dollars instead and refit:

```
> g <- lm(sr ~ pop15+pop75+I(dpi/1000)+ddpi, savings)
> summary(g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   28.566      7.355    3.88  0.00033
pop15         -0.461      0.145   -3.19  0.00260
pop75         -1.691      1.084   -1.56  0.12553
I(dpi/1000)   -0.337      0.931   -0.36  0.71917
ddpi          0.410      0.196    2.09  0.04247
Residual standard error: 3.8 on 45 degrees of freedom
Multiple R-Squared:  0.338, Adjusted R-squared:  0.28
F-statistic: 5.76 on 4 and 45 DF, p-value: 0.00079
```

What changed and what stayed the same?

One rather thorough approach to scaling is to convert all the variables to standard units (mean 0 and variance 1) using the `scale()` command:

```
> scsav <- data.frame(scale(savings))
> g <- lm (sr , scsav)
> summary (g) Coefficients:
              Estimate Std Error t value Pr(>|t|)
(Intercept)  4.0e-16      0.1200  3.3e-15   1.0000
pop15        -0.9420      0.2954  -3.19    0.0026
pop75        -0.4873      0.3122  -1.56    0.1255
dpi          -0.0745      0.2059  -0.36    0.7192
ddpi          0.2624      0.1257   2.09    0.0425
Residual standard error: 0.849 on 45 degrees of freedom
Multiple R-Squared: 0.338, Adjusted R-squared: 0.28
F-statistic: 5.76 on 4 and 45 DF, p-value: 0.00079
```

As may be seen, the intercept is zero. This is because the regression plane always runs through the point of the averages, which because of the centering, is now at the origin. Such scaling has the advantage of putting all the predictors and the response on a comparable scale, which makes comparisons simpler. It also allows the coefficients to be viewed as a kind of partial correlation—the values will always be between minus one and one. It also avoids some numerical problems that can arise when variables are of very different scales. The downside of this scaling is that the regression coefficients now represent the effect of a one standard unit increase in the predictor on the response in standard units—this might not always be easy to interpret.

5.3 Collinearity

When some predictors are linear combinations of others, then $X^T X$ is singular, and we have (exact) collinearity. There is no unique least squares estimate of β . If $X^T X$ is close to singular, we have collinearity (some call it multicollinearity). This causes serious problems with the estimation of β and associated quantities, as well as the interpretation. Collinearity can be detected in several ways:

1. Examination of the correlation matrix of the predictors may reveal large *pairwise* collinearities.
2. A regression of x_i on all other predictors gives R_i^2 . Repeat for all predictors. R_i^2 close to one indicates a problem. The offending linear combination may be discovered by examining these regression coefficients.
3. Examine the eigenvalues of $X^T X$, where λ_1 is the largest eigenvalue with the others in decreasing order. Relatively small eigenvalues indicate a problem. The condition number is defined as:

$$\kappa = \sqrt{\frac{\lambda_1}{\lambda_p}}$$

where $\kappa \geq 30$ is considered large. κ is called the condition number. Other condition numbers, $\sqrt{\lambda_1/\lambda_i}$ are also worth considering because they indicate whether more than just one independent linear combination is to blame. Alternative calculations involve standardizing the predictors and/or excluding the intercept term. Collinearity makes some of the parameters hard to estimate. Define:

$$S_{x_j x_j} = \sum_i (x_{ij} - \bar{x}_j)^2$$

then:

$$\text{var } \hat{\beta}_j = \sigma^2 \left(\frac{1}{1 - R_j^2} \right) \frac{1}{S_{x_j x_j}}$$

We can see that if x_j does not vary much, then the variance of $\hat{\beta}_j$ will be large. Another consequence of this equation is that it tells us which designs will minimize the variance of the regression coefficients if we have the ability to place the X . Orthogonality means that $R_j^2 = 0$ which minimizes the variance. Also we can maximize $S_{x_j x_j}$ by spreading X as much as possible. The maximum is attained by placing half the points at the minimum practical value and half at the maximum. Unfortunately, this design assumes the linearity of the effect and would make it impossible to check for any curvature. So, in practice, most would put some design points in the middle of the range to allow checking of the fit.

If R_j^2 is close to one, then the variance inflation factor $\frac{1}{1 - R_j^2}$ will be large.

Collinearity leads to imprecise estimates of β . The signs of the coefficients can be the opposite of what intuition about the effect of the predictor might suggest. The standard errors are inflated so that t-tests may fail to reveal significant factors. The fit becomes very sensitive to measurement errors where small changes in y can lead to large change in $\hat{\beta}$.

Car drivers like to adjust the seat position for their own comfort. Car designers would find it helpful to know where different drivers will position the seat depending on their size and age. Researchers at the HuMoSim laboratory at the University of Michigan collected data on 38 drivers. They measured age in years, weight in pounds, height with shoes and without shoes in cm, seated height arm length, thigh length, lower leg length and hipcenter the horizontal distance of the midpoint of the hips from a fixed location in the car in mm. We fit a model with all the predictors:


```

> data (seatpos)
> g <- lm (hipcenter ~ . , seatpos)
> summary (g)
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  436.4321    166.5716   2.62    0.014
Age           0.7757     0.5703    1.36    0.184
Weight        0.0263     0.3310    0.08    0.937
HtShoes       -2.6924     9.7530   -0.28    0.784
Ht            0.6013    10.1299    0.06    0.953
Seated        0.5338     3.7619    0.14    0.888
Arm          -1.3281     3.9002   -0.34    0.736
Thigh        -1.1431     2.6600   -0.43    0.671
Leg          -6.4390     4.7139   -1.37    0.182
Residual standard error: 37.7 on 29 degrees of freedom
Multiple R-Squared: 0.687, Adjusted R-squared: 0.6
F-statistic: 7.94 on 8 and 29 DF, p-value: 1.31e-05

```

This model already shows the signs of collinearity. The p-value for the F-statistics is very small and the R^2 is quite substantial, but none of the individual predictors is significant. We take a look at the pairwise correlations:

```

> round(cor(seatpos), 3)
      Age Weight HtShoes   Ht Seated   Arm   Thigh
Age      1.000  0.081  -0.079 -0.090 -0.170  0.360  0.091
Weight   0.081  1.000  0.828  0.829  0.776  0.698  0.573
HtShoes  -0.079  0.828  1.000  0.998  0.930  0.752  0.725
Ht       -0.090  0.829  0.998  1.000  0.928  0.752  0.735
Seated   -0.170  0.776  0.930  0.928  1.000  0.625  0.607
Arm       0.360  0.698  0.752  0.752  0.625  1.000  0.671
Thigh    0.091  0.573  0.725  0.735  0.607  0.671  1.000
Leg      -0.042  0.784  0.908  0.910  0.812  0.754  0.650
hipcenter 0.205 -0.640 -0.797 -0.799 -0.731 -0.585 -0.591
      Leg hipcenter
Age      -0.042  0.205
Weight   0.784 -0.640
HtShoes  0.908 -0.797
Ht       0.910 -0.799
Seated   0.812 -0.731
Arm       0.754 -0.585
Thigh    0.650 -0.591
Leg      1.000 -0.787
hipcenter -0.787  1.000

```

There are several large pairwise correlations both between predictors and between predictors and the response. Now we check the eigendecomposition:

```

> x <- model.matrix(g)[-1] >
e <- eigen (t(x) %*% x)
> e$val

```

```
[1] 3.6537e+06 2.14796+04 9.0432e+03 2.9895e + 02 1.48396+02
[6] 8.1174e+01 5.3362e + 01 7.2982e+00 >
sqrt(e$val[1]/e$val)
[1] 1.000 13.042 20.100 110.551 156.912 212.156 261.667
[8] 707.549
```

There is a wide range in the eigenvalues and several condition numbers are large. This means that problems are being caused by more than just one linear combination.

Now check the variance inflation factors (VIFs). For the first variable this is:

```
> summary (lm (x [, 1] ~ x[,-1]))$r.squared
[1] 0.49948
> 1/(1-0.49948)
[1] 1.9979
```

which is moderate in size—the VIF for orthogonal predictors is one. Now we compute all the VIFs in one go, using a function from the *f* araway package:

```
> vif (x)
      Age Weight  HtShoes      Ht Seated      Arm Thigh
1.9979 3.6470 307.4294 333.1378 8.9511 4.4964 2.7629
      Leg
6.6943
```

There is much variance inflation. For example, we can interpret $\sqrt{307.4} = 17.5$ as telling us that the standard error for height with shoes is 17.5 times larger than it would have been without collinearity. We cannot apply this as a correction because we did not actually observe orthogonal data, but it does give us a sense of the size of the effect.

There is substantial instability in these estimates. Measuring the hipcenter is difficult to do accurately and we can expect some variation in these values. Suppose the measurement error had a SD of 10 mm. Let's see what happens when we add a random perturbation of this size to the response:

```
> g <- lm (hipcenter+10*rnorm(38) ~ . , seatpos)
> summary (g)
Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 501.295 164.752 3.04 0.0049
Age          0.632   0.564  1.12 0.2720
Weight       0.129   0.327  0.39 0.6967
HtShoes      -2.329   9.647 -0.24 0.8109
Ht           0.705  10.019  0.07 0.9444
Seated       -0.751   3.721 -0.20 0.8414
Arm          -0.951   3.858 -0.25 0.8070
Thigh        -1.879   2.631 -0.71 0.4808
Leg          -7.087   4.662 -1.52 0.1393
Residual standard error: 37.3 on 29 degrees of freedom Multiple
R-Squared: 0.696, Adjusted R-squared: 0.612
F-statistic: 8.3 on 8 and 29 DF, p-value: 8.71e-06
```

Although the R^2 and standard error are very similar to the previous fit, we see much larger changes in the coefficients indicating their sensitivity to the response values caused by the collinearity.

One cure for collinearity is amputation. We have too many variables that are trying to do the same job of explaining the response. When several variables, which are highly correlated, are each associated with the response, we have to take care that we do not conclude that the variables we drop have nothing to do with the response. Examine the full correlation matrix above. Consider just the correlations of the length variables:

```
> round (cor (x [ , 3:8] ), 2)
```

	HtShoes	Ht Seated	Arm	Thigh	Leg
HtShoes	1.00	1.00	0.93	0.75	0.72
Ht	1.00	1.00	0.93	0.75	0.73
Seated	0.93	0.93	1.00	0.63	0.61
Arm	0.75	0.75	0.63	1.00	0.67
Thigh	0.72	0.73	0.61	0.67	1.00
Leg	0.91	0.91	0.81	0.75	0.65

These six variables are strongly correlated with each other—any one of them might do a good job of representing the other. We pick height as the simplest to measure. We are not claiming that the other predictors are not associated with the response, just that we do not need them all to predict the response:

```
> g2 <- lm (hipcenter ~ Age + Weight + Ht, seatpos)
> summary (g2)
```

Coefficients:

	Estimate	Std. Error	t value	Pr (> t)
(Intercept)	528.29773	135.31295	3.90	0.00043
Age	0.51950	0.40804	1.27	0.21159
Weight	0.00427	0.31172	0.01	0.98915
Ht	-4.21190	0.99906	-4.22	0.00017

Residual standard error: 36.5 on 34 degrees of freedom
 Multiple R-Squared: 0.656, Adjusted R-squared: 0.626
 F-statistic: 21.6 on 3 and 34 DF, p-value: 5.13e-08

Comparing this with the original fit, we see that the fit is very similar in terms of R^2 , but much fewer predictors are used. Further simplification is clearly possible.

If you must keep all your variables in the model, you should consider alternative methods of estimation such as ridge regression.

The effect of collinearity on prediction depends on where the prediction is to be made. The greater the distance is from the observed data, the more unstable the prediction. Distance needs to be considered in a Mahalanobis rather than a Euclidean sense.

Exercises

1. Using the faithful data, fit a regression of duration on waiting. Assuming that there was a measurement error in waiting of 30 seconds, use the SIMEX method to obtain a better estimate of the slope.

2. What would happen if the SIMEX method was applied to the response error variance rather than predictor measurement error variance?
3. Using the divorce data:
 - (a) Fit a regression model with divorce as the response and unemployed, femlab, marriage, birth and military as predictors. Compute the condition numbers and interpret their meaning.
 - (b) For the same model, compute the VIFs. Is there evidence that collinearity causes some predictors not to be significant? Explain.
 - (c) Does the removal of insignificant predictors from the model reduce the collinearity? Investigate.
4. For the longley data, fit a model with Employed as the response and the other variables as predictors.
 - (a) Compute and comment on the condition numbers.
 - (b) Compute and comment on the correlations between the predictors.
 - (c) Compute the variance inflation factors.
5. For the prostate data, fit a model with lpsa as the response and the other variables as predictors.
 - (a) Compute and comment on the condition numbers.
 - (b) Compute and comment on the correlations between the predictors.
 - (c) Compute the variance inflation factors.

